

Role and Contribution of Women in Modern and Ancient India

Gyanvir (BSC Hons Physics) & Yuvraj Kaushik (BCOM
Hons)

Roll No: 26205 & 25410

UNIVERSITY ROLL NO. - 24035567019
and 24035504109

VAC: Science and Society

Introduction

- Women in India have been central to shaping scientific, political, and social foundations since the Vedic era.
- The journey spans from ancient scholars to modern-day entrepreneurs, scientists, and world-class athletes.
- Despite shifting social constraints over centuries, Indian women have consistently displayed leadership and excellence.
- A comparative study highlights the continuity of intellectual strength and progress in nation-building.

Mathematics – Ancient & Modern

- Ancient: Lilavati (12th Century) - Associated with scholar Bhaskaracharya; her namesake text covers arithmetic and algebra.
- Lilavati's work reflects the high level of mathematical proficiency women could attain in ancient India.
- Modern: Shakuntala Devi ('Human Computer') - Famous for her lightning-fast mental calculations globally.
- Devi's books and public demonstrations made mathematics accessible and interesting to millions of people.



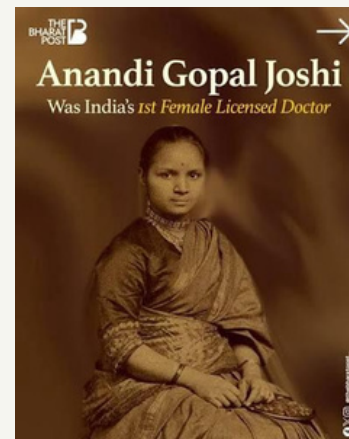
Science – Philosophy & Research

- Ancient: Gargi Vachaknavi - A Vedic philosopher renowned for her role in public debates at King Janaka's court.
- Gargi's inquiries into cosmology and metaphysics displayed an early scientific temperament and rational spirit.
- Modern: Dr. Asima Chatterjee - A pioneer in organic chemistry who focused on medicinal plants of India.
- Her research led to breakthrough drugs for malaria and epilepsy, earning her global recognition.



Medicine – From Ayurveda to Modern Degrees

- Ancient: Maitreyi - A Vedic scholar whose dialogues with Sage Yajnavalkya touched on physiology and ethics.
- Her philosophical contributions laid the groundwork for the holistic health principles seen in Ayurveda.
- Modern: Dr. Anandi Gopal Joshi - Overcame immense social barriers to become India's first female licensed doctor in 1886.
- She studied Western medicine in the US, inspiring generations of Indian women to enter the medical field.



Technology & Space Science

- Ancient: Lopamudra - A prominent figure in Vedic society associated with early structured thinking and natural science.
- Modern: Ritu Karidhal ('Rocket Woman of India') - Played a pivotal role in ISRO's Mars Orbiter Mission (Mangalyaan).
- Modern: Nandini Harinath - Contributed significantly to navigation and trajectory planning for deep space missions.
- These scientists showcase women's leadership in high-stakes, advanced space technology and engineering.



Politics & Governance

- Ancient: Rani Didda - Ruled Kashmir independently for decades, proving mastery in administration and stability.
- Modern: Indira Gandhi - India's first female Prime Minister; introduced bank nationalization and poverty alleviation.
- Modern: Justice Fathima Beevi - The first female judge of the Supreme Court of India, breaking the legal glass ceiling.
- These leaders transformed the political and judicial landscape, emphasizing self-reliance and social justice.



Defense & Physical Prowess

- Ancient: Rani Lakshmibai - The legendary Queen of Jhansi who led her army against British forces in 1857.
- Princess Chitrangada - A figure from the Mahabharata skilled in martial arts, representing ancient physical training.
- Modern: Lt Gen Punita Arora - The first woman in the Indian Armed Forces to reach the rank of Lieutenant General.
- Modern: P.V. Sindhu - Olympic medalist and world champion who brought global glory to Indian badminton.



Social Reform & Business

- Social: Mother Teresa - Dedicated her entire life to serving the poor and sick, establishing global charities.
- Ancient Entrepreneurship: Archaeological evidence from the Indus Valley suggests women engaged in trade and craft.
- Modern Business: Kiran Mazumdar-Shaw - Founder of Biocon, she revolutionized India's biotechnology sector.
- These individuals highlight women's contributions to human welfare and the modern industrial economy.



Impactful Women Scientists

- Janaki Ammal: Pioneering botanist whose work on sugarcane hybrids helped India develop sweet varieties.
- Rajeshwari Chatterjee: The first woman engineer from Karnataka; a global expert in Microwave Engineering.
- Tessy Thomas ('Missile Woman'): Project Director for Agni-IV, securing India's defense technology capabilities.
- Gagandeep Kang: Renowned virologist who played a crucial role in developing the rotavirus vaccine.

Key Contributions of Ancient Scientists

- Mathematics: The concept of Zero (Shunya) as a number and the place-value decimal system originated in India.
- Medicine: Sushruta ('Father of Surgery') performed early plastic surgeries and documented 120+ instruments.
- Astronomy: Aryabhata proposed a spherical Earth that rotates on its axis and accurately explained eclipses.
- Holistic Health: Charaka's 'Charaka Samhita' focused on diet, hygiene, and the three doshas (Vata, Pitta, Kapha).

Modern Scientific Giants

- Sir C.V. Raman: Discovery of the Raman Effect (Nobel Prize 1930) explaining how light scatters.
- Satyendra Nath Bose: Collaborated with Einstein on Bose-Einstein statistics; 'Bosons' are named after him.
- Homi J. Bhabha & Vikram Sarabhai: Foundations of India's nuclear program and space program (ISRO).
- A.P.J. Abdul Kalam: 'Missile Man' who spearheaded the development of India's ballistic missile technology.

Conclusion

- Indian history reflects a long tradition where women's intellectual and leadership roles were vital.
- From philosophical debates in ancient courts to modern space laboratories, the legacy remains unbroken.
- Modern India has expanded these ancient opportunities through constitutional rights and social reform.
- Recognizing these achievements is crucial for building an inclusive and progressive nation.

PYQs

Q1. Discuss the role of Indian scientists, including prominent women scientists, who have made significant contributions to the development of science and technology from ancient to modern times

Ancient Period: Foundations of Logic and Mathematics

- In ancient India, science was deeply intertwined with philosophy and practical necessity.
- Aryabhata (5th Century): The father of Indian mathematics and astronomy. He calculated the value of π (Pi) to four decimal places and proposed that the Earth revolves around the sun.
- Sushruta: Known as the "Father of Surgery." His text, the Sushruta Samhita, describes over 300 surgical procedures, including early forms of rhinoplasty (plastic surgery) and cataract surgery.
- Charaka: The "Father of Ayurveda." His work, the Charaka Samhita, laid the groundwork for ancient Indian medicine, focusing on the balance of the body's three doshas.
- Brahmagupta: The first to provide rules to compute with zero and solved quadratic equations.

2—Modern Period: Physics and Institutional Building

- The 20th century saw Indian scientists gain global recognition through rigorous experimental and theoretical work.
- Sir C.V. Raman: Won the Nobel Prize in Physics (1930) for the discovery of the Raman Effect, which explains how light scatters when passing through a transparent material.
- Satyendra Nath Bose: Collaborated with Albert Einstein to develop Bose-Einstein statistics, which describe the behavior of subatomic particles now known as Bosons.
- Homi J. Bhabha: The architect of India's nuclear program. He founded the Tata Institute of Fundamental Research (TIFR).
- Vikram Sarabhai: The father of the Indian Space Program, instrumental in the establishment of ISRO.
- A.P.J. Abdul Kalam: Known as the "Missile Man of India" for his pivotal role in developing India's ballistic missile and launch vehicle technology.

- 3. Prominent Women Scientists • Women have overcome significant social barriers to lead breakthroughs in various scientific domains.
- •Janaki Ammal: A pioneering botanist who conducted major research on cytogenetics and phytogeography. Her work on sugarcane hybrids helped India develop indigenous sweet varieties.
- •Asima Chatterjee: Renowned for her work in organic chemistry. She developed anti-epileptic and anti-malarial drugs from medicinal plants and was the first woman to receive a Doctorate of Science from an Indian university.
- •Rajeshwari Chatterjee: The first woman engineer from Karnataka, she made significant contributions to the field of Microwave Engineering and antennae.
- •Tessy Thomas: Known as the "Missile Woman of India," she served as the Project Director for the Agni-IV missile, making her a key figure in India's defense technology.
- •Gagandeep Kang: A leading virologist and the first Indian woman to be elected as a Fellow of the Royal Society. She played a crucial role in developing the rotavirus vaccine.

Q2.Describe key contributions made by ancient Indian scientists in fields such as mathematics, astronomy and medicine

- Ancient India was a global hub for intellectual pursuit, where science and logic were used to solve practical problems and understand the cosmos. Here is a description of the key contributions in mathematics, astronomy, and medicine.
- 1. Mathematics: The Language of Numbers
 - Ancient Indian mathematicians laid the foundation for the modern numbering system used worldwide today.
 - The Concept of Zero: Perhaps the most significant contribution was the definition of zero (Shunya) as a number rather than just a placeholder. Brahmagupta (7th Century) was the first to establish mathematical rules for computing with zero.
 - Decimal System: The place-value system and the decimal notation (base-10) originated in India, which simplified complex calculations significantly compared to the Roman numeral system.
 - Geometry (Sulba Sutras): These ancient texts (c. 800–500 BCE) contain early formulations of the Pythagorean Theorem and methods for constructing geometric shapes for sacrificial altars.
 - Algebra and Trigonometry: Aryabhata introduced the concept of sine (jya) and calculated the value of π to be approximately 3.1416.
- 2. Astronomy: Understanding the Heavens
 - Indian astronomy was highly advanced, often using mathematical models to predict celestial events without the aid of telescopes.
 - Heliocentric Ideas: In his work *Aryabhatiya*, Aryabhata proposed that the Earth is spherical and rotates on its own axis. He also correctly explained that eclipses are caused by shadows cast by the Earth and the Moon, rather than mythological demons.